



Research paper



Impact of using a different dispatch strategy in the operating phase than in the design phase for decentralized energy systems

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ABSTRACT

Innovative dispatch strategies are essential for renewable energy systems to ensure optimal performance, cost-effectiveness, and sustainability. Optimal dispatch is often assumed to design renewable energy systems, while rule-based dispatch strategies are usually applied in the operating phase. This study examines the effects of using a different dispatch strategy in the operating phase than in the design phase. For this purpose, a mathematical optimization model is used to design an energy system with two rule-based dispatch strategies and the optimal dispatch to evaluate their effects on system costs, component dimensioning, and selection. Three case studies were used to show that, firstly, the system cost, component dimensioning, and component selection change depending on the dispatch strategies used in the design phase. Second, we show that the planned costs increase by up to 10.54 % if the system is designed with an optimal dispatch but a rule-based strategy is used in the operating phase. Third, we show that the system cost would be up to 6.07 % lower if the correct rule-based strategy had been directly applied in the design phase. Furthermore, in the multi-energy system, with additional consideration of the heat demand, it was even found that the design with optimal dispatch can lead to undersized heat supply components, resulting in unreliable heat supply when a rule-based dispatch strategy is used in the operating phase. These findings highlight the critical impact of dispatch strategies on the economic and operational feasibility of renewable energy projects, emphasizing the need for integrated planning across both phases. This is crucial for optimizing system performance and cost-effectiveness. We recommend that engineers and planners consider these insights to enhance the reliability and efficiency of renewable energy system planning.

1. Introduction

The subject of the present paper is the design of decentralized energy systems (DES) based on renewable energy sources, with a particular focus on the dispatch strategy and its impact on the design and operating phase. While conventional technologies have been used to meet energy needs in the past, DES based on renewable energy sources will play an increasingly important role in the future energy system (Triebel et al., 2022). Renewable energy systems require flexible components to balance the temporal gap between inflexible supply and demand. For this purpose, dispatchable energy converters, energy storage systems, or flexible energy demands are needed. Innovative dispatch strategies are required to efficiently dispatch these flexible components and ensure a reliable energy supply (Kratochwill et al., 2020).

During the design phase of DES, component dimensioning and selection must be done, requiring the consideration of a specific dispatch strategy. There are two fundamentally different methodologies

for component dimensioning and dispatch: Using expert knowledge to formulate heuristic rules or using optimization models to compute an optimal solution. Research in the design of decentralized energy systems has increased massively in recent years. The present paper focuses on rules-based approaches – as they are often used in the operation of plants – and optimization approaches – as these are usually applied in the literature to design energy systems. Table 1 lists some relevant publications for possible combinations of the methodological approaches to design decentralized energy systems without any claim to completeness. For other approaches (e.g., model predictive control or methods of machine learning), the reader is referred to meta-studies for more in-depth information on this topic (Taghizadegan Kalantari et al., 2018; Hannan et al., 2020; Abdmouleh et al., 2017; Tarraq et al., 2021; Cuisinier et al., 2021).

Simplified heuristic rules are often used to roughly design energy systems without high computational effort (first column in Table 1).

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Nomenclature

List of Symbols

A_{HS}^{sur}	Surface area of heat storage	u_{PV}	Normalized input profile
a_i	Annuity payment factor	V_{HS}	Heat storage tank volume
C_B^r	Rated battery energy capacity	w_p	Number of represented periods
c_p	Specific heat capacity	z_{ss}	Self-sufficiency
c_i	Specific cost/price	Z_i	State of storage i
E	Energy	α_{PV}	Module azimuth angle
f	energy flow direction factor	γ_{PV}	Module tilt angle
F_B	Battery state of energy	η	Efficiency
h_{HS}	Height of storage tank	ϵ_{HP}	Efficiency factor of heat pump
j_c	Imputed interest rate	Π_i	Specific power of storage i
J	Annual cost	ρ_{med}	Density of medium
k_B^{sd}	Self-discharge rate constant	Component Indices	
l_{PV}	Photovoltaic system loss factor	B	Battery storage
P	Electrical power	EB	Electric boiler
$\dot{Q}$	Heat flow	ED	Electrical demand
r_{HS}	Radian of heat storage	EG	Electrical grid
S_i	Dimensioning variable of comp. i	HP	Heat pump
T_i	Life span of component i	HS	Heat storage
T	Temperature	PV	Photovoltaic system
t	Time		

In Orth et al. (2022), a simple rule of thumb is developed for dimensioning residential battery systems based on the size of the associated photovoltaic system. In Vandewalle and D'haeseleer (2014) and Cardona and Piacentino (2003), aggregated load duration curves are used to dimension combined heat and power units (CHP), assuming a priori either heat or electricity-driven operation. There are plenty of additional examples, mainly from manufacturers (Viessmann Deutschland GmbH, 2014; Reflex Winkelmann GmbH, 2019), and guidelines (V.D.I. - The Association of German Engineers, 2023, 2013) for designing components using straightforward rules. However, these rules cannot adequately account for local conditions, dispatchable components, and combinations of technologies, thereby only roughly realizing the potential of DES based on renewable energy sources.

Mathematical models are often used in the scientific literature to design DES to determine a cost-optimal dimensioning (second column in Table 1). Within these energy system models, either rule-based dispatch strategies or optimized dispatch strategies can be applied. Rule-based dispatch strategies rely on predefined rules and heuristics developed through expert knowledge to decide which component should generate or consume and determine the concrete value of the power. These strategies are straightforward and computationally efficient but may not fully use the system's potential due to their simplistic nature. In contrast, optimization-based dispatch strategies determine the most efficient way to operate energy system components. These strategies aim to minimize costs or maximize performance by solving complex optimization problems, often requiring significant computational resources and are limited by assuming perfect forecasts.

Rule-based dispatch strategies are often used in the design of conventional energy systems. For example, CHPs are usually dimensioned, assuming either heat- or electricity-driven operation (e.g., Li et al., 2018; Urbanucci and Testi, 2018) or full-load operation (e.g., Caliano et al., 2017). Wilke et al. (2018) formulates a ranking list to dispatch the components in a renewable energy system. Other authors,

such as Simeoni et al. (2018) or Wang et al. (2020), use a unique formulation-based dispatch strategy to design a more complex energy system. In addition to clearly formulated rules, fuzzy rules (e.g., Perera et al., 2017) or neural networks (e.g., Perera et al., 2020) are also used as a form of rule-based dispatch in design optimization.

Most researchers optimize system design and dispatch simultaneously (last row in Table 1). A widely used approach is optimization using Mixed Integer Linear Programming (MILP) (e.g., Gabrielli et al., 2018; Wouters et al., 2017; Yang et al., 2015). The authors in Gabrielli et al. (2018) and Wouters et al. (2017) using this approach to select and size system components of a renewable-based multi-energy system for a neighborhood. Yang et al. (2015) additionally optimize the type, number, and location of equipment through a superstructure-based approach. Other authors decompose the problem into a two-stage optimization approach (e.g., Elsidio et al., 2017; Schmeling et al., 2022; Luo et al., 2020). For instance, Elsidio et al. (2017) employs an evolutionary algorithm to select and size units in the upper level. Meanwhile, the optimal dispatch problem is linearized and optimized at the lower level using a commercial MILP solver. Luo et al. (2020) uses this approach to show that optimal dispatch results in a 14 % reduction in year-round biomass consumption compared to a design using conventional dispatch strategies. Similar results show Lee and Kum (2019). When applied, the design with optimal dispatch leads to a 5 – 10 % cost reduction compared to a design with rule-based dispatch.

The influence of dispatch strategy on optimal design has already been considered in the literature (e.g., Schmeling et al., 2022; Luo et al., 2020; Lee and Kum, 2019). However, a significant research gap remains in understanding the consequences of applying different dispatch strategies in the design and operating phases. Specifically, while many studies focus on the benefits of optimal dispatch during the design phase, they often overlook the practical reality that rule-based strategies are frequently employed during the operating phase.

Table 1

Overview of different application examples for dimensioning and dispatching an energy system in the design phase.

		Dimensioning method	
		Heuristic rule	Optimization
	Heuristic rule	e.g. Orth et al. (2022), Vandewalle and D'haeseleer (2014), Cardona and Piacentino (2003), Viessmann Deutschland GmbH (2014), Reflex Winkelmann GmbH (2019), V.D.I. - The Association of German Engineers (2023, 2013)	e.g. Li et al. (2018), Urbanucci and Testi (2018), Caliano et al. (2017), Simeoni et al. (2018), Wilke et al. (2018), Wang et al. (2020), Perera et al. (2017, 2020)
Dispatch method	Optimization	–	e.g. Gabrielli et al. (2018), Wouters et al. (2017), Yang et al. (2015), Elsidio et al. (2017), Schmeling et al. (2022), Luo et al. (2020), Lee and Kum (2019)

This discrepancy can lead to suboptimal system performance and increased costs. Addressing this gap is crucial for providing actionable insights that ensure renewable energy systems are economically and operationally efficient across their lifecycle.

The present work aims to investigate the impact of applying different dispatch strategies in the design phase and to understand how the costs planned in the design phase increase when a different dispatch strategy is used in the operating phase. The following aspects are considered: (1) Comparison of the design using rule-based dispatch with the design using the optimal dispatch. (2) Analysis of the impact of using an optimal dispatch strategy in the design phase followed by a rule-based dispatch strategy in the operating phase on system costs. (3) Determination of the cost savings if the dispatch strategy used in the operating phase had been applied directly in the design phase. These objectives are investigated in a model-based analysis of three detailed case studies.

The paper is structured as follows: In Section 2, the methodology describes the mathematical model of a simple energy system and the different dispatch strategies applied in the study. The mathematical model is used in Section 3 within two case studies. Therein, the results of the optimized dimensioning for the design phase and the simulated results of the subsequent operating phase are shown and discussed. The analysis is then transferred to a third case study considering a multi-energy system with a heat supply in Section 4. Finally, the conclusions and limitations are presented in Section 5.

2. Model description for simple energy system

This section describes the mathematical model for the simple energy system, including information on the architecture of the analyzed system (Section 2.1), the formulation of the design optimization problem (Section 2.2), the component models (Section 2.3) and the different dispatch strategies under consideration (Section 2.4). The following mathematical model is implemented in the equation-based modeling language *Modelica* (Open Source Modelica Consortium, 2022).

2.1. System architecture

Fig. 1 shows the topology of the considered energy system. A rooftop photovoltaic system (P_{PV}), lithium-ion battery storage ($P_B > 0$ charging; $P_B < 0$ discharging), and a public grid connection (P_{EG}) meet the electrical demand (P_{ED}). Due to the grid connection, purchasing ($P_{EG} > 0$) and feeding ($P_{EG} < 0$) electrical energy is possible. According to the typical price and fee system, the price of purchasing electrical

power from the public grid is divided into a peak power charge c^{power} and an energy demand charge c^{energy} . For this reason, battery storage is used, on the one hand, to balance the temporal gap between the fluctuating demand and the renewable supply and, on the other hand, to reduce the maximum grid power.

2.2. System design optimization

The system design optimization problem is explained step by step in the following section.

$$\text{minimize}_{S_i, F_{B,\text{start}}} J_c + J_m + J_o \quad \forall i \in \{PV, B\}, \quad (1)$$

$$\text{subject to} \quad \sum_{i \in I} f_i \cdot P_i(t) = 0 \quad \forall t, \quad (2)$$

$$P_{EG}(t) \leq P_{EG}^{\text{max}} \quad \forall t, \quad (3)$$

$$F_{B,\text{end}} \geq F_{B,\text{start}}, \quad (4)$$

$$\text{eqs. (5) – (23)}$$

$$\text{with} \quad I \in \{ED, PV, B, EG\}.$$

The dimensioning variables S_i of the components (photovoltaic system's rated power P_{PV}^{rated} and battery storage's rated capacity C_B^{rated}) and the battery's initial SOE $F_{B,\text{start}}$ are varied to minimize the TAC. The TAC is given by three contributions: Annual capital J_c , maintenance J_m , and operation cost J_o . The annual capital costs are calculated using the annuity method:

$$J_c = \sum_{i \in \{PV, B\}} S_i \cdot c_i^{\text{acq}} \cdot a_i, \quad (5)$$

where S_i is the component dimensioning variable, c_i^{acq} is the specific acquisition cost, and a_i is the annuity payment factor. Assuming an imputed interest rate j_c of 3.5 % and a component-dependent life span T_i , the annuity payment factor is calculated as follows:

$$a_i = \frac{(1 + j_c)^{T_i} \cdot j_c}{(1 + j_c)^{T_i} - 1} \quad \forall i \in \{PV, B\}. \quad (6)$$

The annual maintenance cost $J_{m,i}$ of component i can then be calculated as a fraction c_i^m of the annual capital cost $J_{c,i}$:

$$J_m = \sum_{i \in \{PV, B\}} J_{c,i} \cdot c_i^m. \quad (7)$$

The parameters for the components' specific acquisition cost, maintenance cost, and life span are listed in Table A.7.

In the present paper, the computation time of the optimization is reduced by using the method of Kotzur et al. (2018) to calculate a set

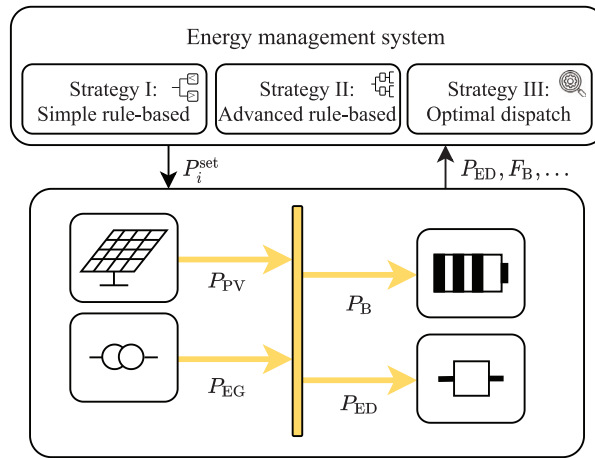


Fig. 1. Topology of the considered energy system: Electrical demand (ED), photovoltaic system (PV), lithium-ion battery (B), electrical grid connection (EG).

of typical representative periods and subsequently utilize them in the analysis to approximate the annual operation cost:

$$J_o = \sum_{p \in \mathbb{P}} J_{o,p} \cdot w_p \quad (8)$$

where w_p is the number of periods represented by period p in the annual profile (hence $\sum_{p \in \mathbb{P}} w_p = 365$), and $J_{o,p}$ are the operation costs in period p . The operation costs of each period

$$J_{o,p} = \int \underbrace{c^{\text{energy}} \cdot P^{\text{purch}}}_{\text{Energy purchasing cost}} - \underbrace{r^{\text{fi}} \cdot P^{\text{fi}}}_{\text{Feed in revenues}} + \underbrace{c^{\text{ss}} \cdot (P_{\text{PV}} - P^{\text{fi}})}_{\text{Self-supply cost}} dt_p + \underbrace{c^{\text{power}} \cdot P_{\text{EG}}^{\text{max}}}_{\text{Peak import cost}} \quad \forall p \in \mathbb{P} \quad (9)$$

are calculated based on the energy cost of the imported electricity

$$P^{\text{purch}} = P_{\text{EG}} \cdot H(P_{\text{EG}}), \quad (10)$$

the revenues for feed-in power

$$P^{\text{fi}} = -P_{\text{EG}} \cdot H(-P_{\text{EG}}), \quad (11)$$

the cost for the renewable self-supply and the charge for the maximum grid power $P_{\text{EG}}^{\text{max}}$. H is the Heaviside function that considers only the positive or negative aspects. The weight factors w_p for each typical period, as well as the energy demand charge price c^{energy} and the peak demand charge price c^{power} , can be found in the appendix for the selected case studies. The feed-in and self-supply prices (r^{fi} and c^{ss}) are according to the current German market regulation 4.46 ct/kWh resp. 2.26 ct/kWh (Federal Government Germany, 2016).

The optimization problem must satisfy the following constraints: Eq. (2) states the continuous power balance to ensure a reliable power supply. The power flow factor f_i describes whether the component i consumes ($f_i = -1$) or supplies energy ($f_i = +1$). Constraint (3) ensures that the grid consumption does not exceed the maximum grid power. Eq. (4) defines the cyclic boundary condition for the battery's SOE. Finally, compliance with the system component models and dispatch strategy must be given.

To better analyze the impact of the dispatch strategies, the degree of self-sufficiency is introduced as follows:

$$z_{\text{ss}} = \sum_{p \in \mathbb{P}} \left(1 - \int \frac{P^{\text{purch}}}{P_{\text{ED}}} dt_p \right) \cdot \frac{w_p}{365}. \quad (12)$$

The optimization is performed with the *particle swarm optimization (generational)* algorithm (GPSO) from the *python parallel optimization library* framework (pygmo) (Biscani and Izzo, 2010).

2.3. System components

The following section presents the input profiles and the component models, including the physical behavior and the operating limits. The physical parameters of the component models are listed in Table A.8.

2.3.1. Electrical demand

The electrical demand profile is adapted for a woodworking company in Germany as an example (Huber et al., 2019). The company consumes about 557 MWh/a of electrical energy with a peak demand of 303 kW ($P_{\text{ED}}^{\text{max}}$). Fig. 2 shows the original annual time series (a) and the calculated reduced time series, including the five typical days (b). The reduced profile includes four typical days and one peak demand day (Day 4).

2.3.2. Photovoltaic system model

The method of Pfenninger and Staffell (2016) is used to calculate the electrical power of the photovoltaic system:

$$P_{\text{PV}} = u_{\text{PV}} \cdot P_{\text{PV}}^{\text{r}}, \quad (13)$$

where P_{PV}^{r} is the rated power of the photovoltaic system, and u_{PV} is the normalized power of the photovoltaic system calculated in advance. Fig. 3 shows the original annual time series (a) and the computed typical days (b).

2.3.3. Battery storage model

A technology-flexible storage model is used to describe the battery storage (Tiemann et al., 2020). The energy balance determines the battery's SOE F_B :

$$C_B^{\text{r}} \cdot \frac{dF_B}{dt} = -k_B^{\text{sd}} \cdot F_B \cdot C_B^{\text{r}} + P_B \cdot \begin{cases} \eta_B & \text{if } P_B \geq 0 \\ \frac{1}{\eta_B} & \text{if } P_B < 0 \end{cases} \quad (14)$$

The rated capacity C_B^{r} is the dimensioning variable of the battery, η_B is the average charge, respectively discharge efficiency, and k_B^{sd} is the self-discharge rate constant. The following rules are given to respect the charge and discharge limits

$$P_B^{\text{max}} = C_B^{\text{r}} \cdot \Pi_B, \quad (15)$$

of the battery and to prevent the battery from deep discharge and overcharging:

$$P_B = \begin{cases} P_B^{\text{max}} & \text{if } 0 < F_B < 1 \wedge P_B^{\text{set}} > P_B^{\text{max}} \\ P_B^{\text{set}} & \text{if } 0 < F_B < 1 \wedge -P_B^{\text{max}} \leq P_B^{\text{set}} \leq P_B^{\text{max}} \\ -P_B^{\text{max}} & \text{if } 0 < F_B < 1 \wedge P_B^{\text{set}} < -P_B^{\text{max}} \\ 0 & \text{otherwise.} \end{cases} \quad (16)$$

The dispatch strategy calculates the battery setpoint power P_B^{set} , defined in the next section.

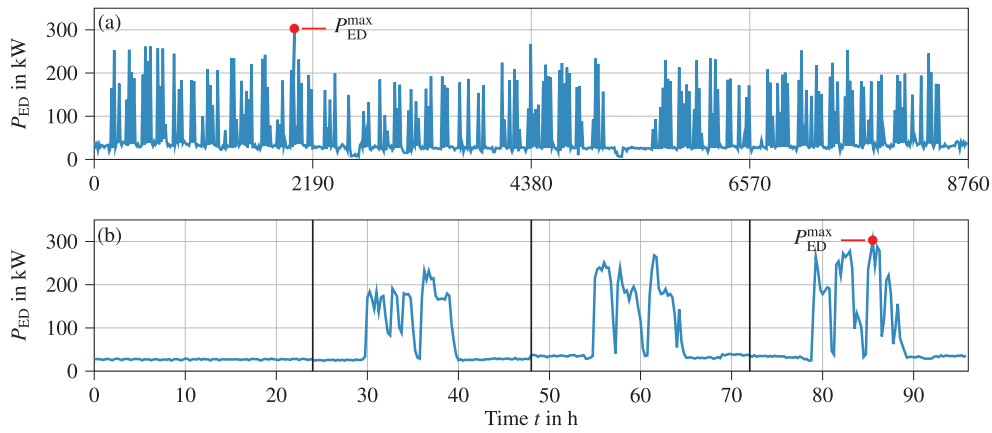


Fig. 2. The original annual profile for the electrical demand (a) and the corresponding, calculated four typical periods (b). The last typical period takes into account the maximum power demand $P_{ED}^{\max}$ of the original profile.

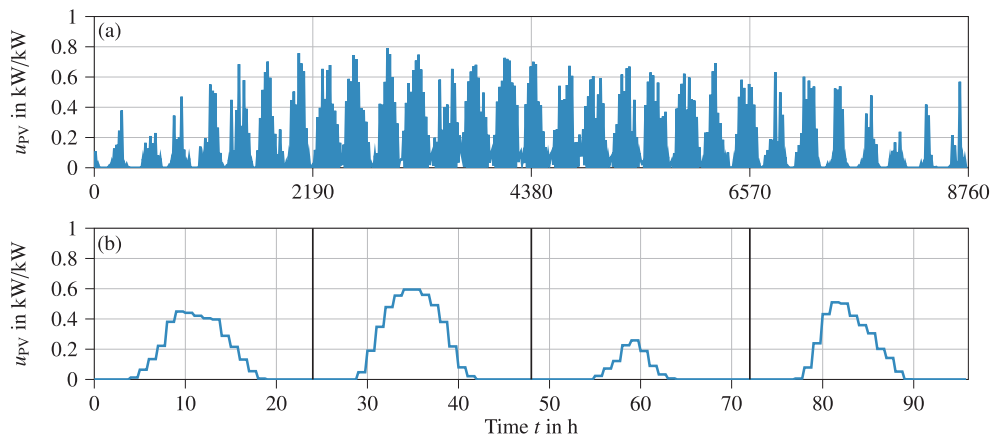


Fig. 3. The original annual time series (a) and the calculated design days (b) for the normalized power of a photovoltaic system located in Frankfurt, Germany.

2.4. Dispatch strategies

The following section introduces three different dispatch strategies to operate the dispatchable components of the energy system (battery storage). First, two rule-based dispatch strategies are introduced, and then an optimization problem for calculating the optimal dispatch is presented. The power of the non-dispatchable components (the electrical demand and the photovoltaic system) is calculated by the component models, and the imported or exported grid power is followed by Eq. (2).

2.4.1. Strategy I: Simple rule-based

Dispatch strategy I aim to minimize the energy purchased from the public grid. Therefore, the battery storage balances the inflexible power difference

$$P_{\text{inflex}} = P_{\text{PV}} - P_{\text{ED}} \quad (17)$$

between the electrical demand and the renewable supply:

$$P_B^{\text{set}} = P_{\text{inflex}}. \quad (18)$$

2.4.2. Strategy II: Advanced rule-based

Dispatch strategy II can serve two purposes by introducing a battery operation mode Z_B . As in strategy I, in mode one ($Z_B = 1$), the battery balances the time gap between the inflexible demand and supply. In contrast, the battery reduces the maximum grid power in mode two

($Z_B = 2$). Switching between the operating modes is done via the battery's desired SOE F_B^{des} :

$$Z_B = \begin{cases} 2 & \text{if } F_B < F_B^{\text{des}} \\ 1 & \text{if } F_B \geq F_B^{\text{des}} \end{cases} \quad (19)$$

The following rule then calculates the battery setpoint power:

$$P_B^{\text{set}} = \begin{cases} P_{\text{inflex}} + P_{\text{EG}}^{\max} & \text{if } Z_B = 2 \\ P_{\text{inflex}} + P_{\text{EG}}^{\max} & \text{if } Z_B = 1 \wedge P_{\text{EG}}^{\max} + P_{\text{inflex}} < 0 \\ P_{\text{inflex}} & \text{if } Z_B = 1 \wedge P_{\text{inflex}} > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (20)$$

Using this dispatch strategy, the battery's desired SOE F_B^{des} is also considered an optimization variable in addition to the introduced optimization variables.

2.4.3. Strategy III: Optimal dispatch

The optimal dispatch problem for calculating the optimal trajectory of the components that can be used (only the battery in this case) is described below.

$$\underset{P_B, P_{\text{EG}}}{\text{minimize}} \quad J_o \quad (21)$$

$$\text{subject to} \quad 0 \leq F_B(t) \leq 1 \quad \forall t, \quad (22)$$

$$-P_B^{\max} \leq P_B(t) \leq P_B^{\max} \quad \forall t, \quad (23)$$

eqs. (2) – (15),

where the battery power is the optimization variable, and the objective is to minimize annual operating costs. In addition to the constraints

Table 2

(a) Component dimensioning S_i and the associated TAC of the considered energy system using the different dispatch strategies (DS) in the design phase (DP). (b) The resulting TAC resulting from applying the different dispatch strategies on the pre-designed systems in the operating phase (OP).

(a) Design phase (DP)			
	Dispatch strategy in DP		
	I	II	III
Dimensioning			
P_{PV} in kW	236.56	201.92	310.78
C_B^r in kWh	0	48.20	191.44
TAC in T€/a	108.80	106.58	104.40
(b) Operating phase (OP)			
	Dispatch strategy in DP		
	I	II	III
	TAC in T€/a		
DS in OP			
I	108.80	110.46	115.40
II	108.80	106.58	108.76
III	108.80	105.68	104.40

from the design optimization (cf. Eqs. (2) and (4)), the operating limits for the battery's SOE (cf. Eq. (22)) and power (cf. Eq. (23)) are formulated as inequality constraints instead of explicit rule-based equations (cf. Eq. (16)).

The optimal dispatch problem is embedded in the overlaid design optimization problem using the *pygmo* framework and solved with the *interior point optimization* algorithm (IpOpt) (Wächter and Biegler, 2006).

3. Results for simple energy system

Two cases are analyzed for the simple energy system to evaluate the impact of the dispatch strategies on the component dimensioning in the design phase and the effect of using a different dispatch in the operating phase.

3.1. Case study 1a: Energy demand and peak power charge

Case study 1a considers a peak power charge and an energy demand charge ($c^{\text{power}} = 102.54 \text{ EUR/kW}$; $c^{\text{energy}} = 15.32 \text{ ct/kWh}$). First, the results of using different dispatch strategies in the design phase are discussed. Then, the effect of using a different dispatch strategy in the operating phase is analyzed based on the system designs provided in advance.

Design phase - Table 2 (a) shows the optimum component dimensioning and the associated TAC applying the different dispatch strategies. Table 3 shows the corresponding level of self-supply z_{ss} and the maximum power purchased from the grid P_{EG}^{max} . The photovoltaic system dimensioning is not significantly affected by the dispatch strategy. At the same time, the rated capacity of the battery storage and the TAC are considerably different for the three different dispatch strategies. As expected, the optimal dispatch (Strategy III) yields the best objective value. This strategy makes it possible to reduce the maximum grid power to 127.11 kW and simultaneously reduce the purchased energy, resulting in a self-sufficiency level of 60.46 %. Due to the resulting higher economic incentive, the battery storage is designed with a higher capacity than applying the two rule-based strategies.

Comparing the two rule-based strategies, the resulting TAC is lower with Dispatch Strategy II than with Strategy I. The strategy I only aims to increase PV self-supply and thus reduce the amount of energy purchased from the public grid (cf. Section 2.4.1). In this scenario, however, the energy demand charge is comparatively low, so it is not worth installing a battery storage system applying this dispatch strategy. Dispatch Strategy II, on the other hand, additionally seeks

Table 3

Maximum grid power P_{EG}^{max} and self-sufficiency z_{ss} using the different dispatch strategies (DS) in the operating phase (OP) with System Design I–III.

	System design		
	I	II	III
DS in OP	P_{EG}^{max} in kW		
I	246.66	249.78	240.01
II	246.66	201.92	131.54
III	246.66	201.53	127.11
	z_{ss} in %		
I	40.26	40.84	60.68
II	40.26	37.07	46.07
III	40.26	40.69	60.46

to minimize the peak power charge (cf. Section 2.4.2). It is, therefore, worth installing a battery storage system with a capacity of 48.20 kWh using this dispatch strategy. This allows the reduction of the maximum grid power to 201.92 kW and a self-supply level of 37.07 % to be achieved simultaneously.

The results show that the choice of dispatch strategy affects both the dimensioning and the selection of components in the energy system. Furthermore, a dispatch strategy better suited to the energy system significantly reduces the TAC. In the following, the systems according to Dispatch Strategies I, II, and III are called System Design I, II, and III, respectively.

Operating phase - Table 2 (b) shows the TAC of the pre-designed systems using the presented dispatch strategies in the operating phase. Since System Design I does not include battery storage, all dispatch strategies lead to the same TAC. The results for System Designs II and III differ from this. As in the design optimization, the optimal dispatch also leads to the best results in the operating phase for each system design, followed by Strategy II. At the same time, it is clear that the planning costs can increase considerably if optimal dispatch (Strategy III) was used in the design phase and rule-based dispatch is applied in the operating phase. The costs increase by 10.54 % resp. 4.18 % for rule-based strategy I resp. II (in System Design III).

Comparing the results of Table 2 (b) on a line-by-line basis shows the impact of how much system costs could be saved if the dispatch strategy used in the operating phase was directly applied in the design phase. For Dispatch Strategy I, this means that if the system were not designed with optimal dispatch but directly with Strategy I, the system costs would be up to 6.07 % cheaper. This effect is the same but less pronounced for Dispatch Strategy II (2.04 %). This is because of two facts. First, Dispatch Strategy II can utilize the battery storage almost as well as the optimal dispatch. Second, the objective function of the design optimization problem for Dispatch Strategy II is relatively flat near the optimum: Changing the dimension of the battery storage will not significantly affect the objective value near the optimum, because with the larger sized battery the peak power can be reduced more (from 201.92 kW to 131.54 kW).

This analysis highlights the following: First, the dispatch strategy used in the design phase will affect the system design and cost. Second, using a different dispatch strategy in the operating phase than in the design phase may result in a suboptimal system design and increased planning costs. In particular, if an optimal dispatch is assumed in the design phase but not realized in the operation phase, the energy system may be suboptimally designed, and increased costs must be expected during operation. Significant cost savings can be made if a realizable dispatch strategy is adopted directly in the design phase.

3.2. Case study 1b: Energy demand charge only

In case study 1b, the peak power charge is neglected while a higher energy demand charge is considered ($c^{\text{power}} = 0 \text{ EUR/kW}$; $c^{\text{energy}} =$

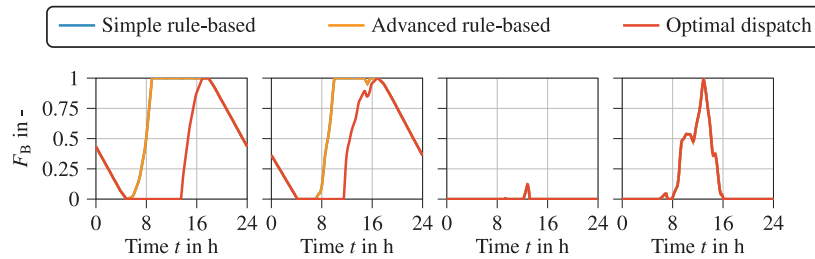


Fig. 4. The course of the battery SOE level (F_B) applying the different dispatch strategies on System Design I.

Table 4

(a) Component dimensioning S_i and the associated TAC of the considered energy system using the different dispatch strategies (DS) in the design phase (DP). (b) The TAC resulting from applying the different dispatch strategies on the pre-designed systems in the operating phase (OP).

(a) Design phase (DP)			
	Dispatch strategy in DP		
	I	II	III
Dimensioning			
P_{PV}^r in kW	530.37	530.37	531.93
C_B^r in kWh	306.36	306.36	306.36
TAC in T€/a	102.937	102.937	102.915
(b) Operating phase (OP)			
	Dispatch strategy in DP		
	I	II	III
	TAC in T€/a		
DS in OP			
I	102.937	102.937	102.938
II	102.937	102.937	102.938
III	102.916	102.916	102.915

21.19 ct/kWh). Consequently, the only economic incentive for battery storage is to balance the temporal gap between the renewable supply and the demand. The analysis follows the same structure as the first case study and is divided into the design and operating phases.

Design phase - Table 4 (a) shows the dimensioning and the associated TAC using the different dispatch strategies. The system is similarly dimensioned for the three different dispatch strategies, and almost the same TAC arises. Only the TAC using the optimal dispatch is slightly lower. The reason for this can be seen in Fig. 4. The Figure shows the resulting charge level curve for the various operating modes using System Design II. The following difference can be seen, particularly in the first and second periods: The application of an optimal dispatch assumes a perfect forecast for all load profiles, allowing the battery storage to be charged at the end of the renewable energy surplus period, thus minimizing the losses due to self-discharge. The two rule-based strategies do not include forecasting. Therefore, they charge the battery at the beginning of the renewable surplus period. For this reason, with optimal dispatch, self-discharge losses are lower, and the economic incentive of the battery storage is slightly higher.

Operating phase - As seen in the design phase, dispatch strategies have a negligible effect. Only the optimal dispatch performs slightly better than the rule-based strategies. Therefore, applying a dispatch strategy different from the one used in the design phase has no significant effect on the TAC.

This case study shows a design optimization problem where the design is independent of the dispatch strategy, so considering a realizable dispatch strategy already in the design phase is irrelevant.

4. Transfer to a multi-energy system

This section analyzes the impact of the dispatch strategy on a multi-energy system using the same procedure as before. First, the updated

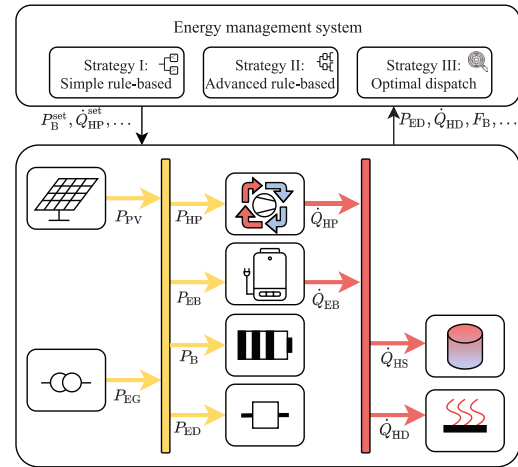


Fig. 5. Topology of the considered energy system with an electric (ED) and a heat demand (HD), a photovoltaic system (PV), a connection to the public grid (EG), an air source heat pump (HP), an electric boiler (EB), a lithium-ion battery (B) and a heat storage (HS).

system is presented in Section 4.1, and then the dispatch strategies are introduced in Section 4.2. Finally, the results of applying the different dispatch strategies in the design and operating phases are detailed in Section 4.3. The additional component models and the updated system parameters can be found in the appendix.

4.1. Updated system

Fig. 5 presents the energy system architecture under investigation. It comprises a small residential district's electrical and thermal ($\dot{Q}_{HD}$) energy demands. Again, the electrical power can be supplied through a photovoltaic system, a lithium-ion battery, or a connection to the public grid. An air source heat pump ($\dot{Q}_{HP}$), an electric boiler ($\dot{Q}_{EB}$), and a heat storage (charging: $\dot{Q}_{HS} > 0$; discharging: $\dot{Q}_{HS} < 0$) are considered for heat supply. The electric boiler and the heat pump increase the fixed electric demand by using electricity (P_{HP} , P_{EB}) to provide heat. The design optimization problem of the extended system is formulated according to that of the simple energy system and can be found in the appendix (cf. Appendix A).

4.2. Dispatch strategies for multi-energy system

In this energy system, the dispatch strategy must specify the heat flow setpoint for the heat pump $\dot{Q}_{HP}^{set}$, electric boiler $\dot{Q}_{EB}^{set}$, and thermal storage $\dot{Q}_{HS}^{set}$ next to the battery setpoint power. This section introduces two rule-based strategies and the optimal dispatch problem.

Table 5

Definition of transition criteria for the state Z_j of component j using the storage water temperature (T_{HS}) and the defined temperature limits for the component j ($T_j^{\text{on}}, T_j^{\text{off}}$).

	pre	$Z_j = 0$	$Z_j = 0.5$	$Z_j = 1$
next				
$Z_j = 0$			$T_{HS} > T_j^{\text{off}}$	
$Z_j = 0.5$				$T_{HS} > T_j^{\text{on}}$
$Z_j = 1$	$T_{HS} < T_j^{\text{on}}$	$T_{HS} < T_j^{\text{on}}$		

4.2.1. Strategy I: Simple rule-based

Strategy I extends the ranking list from Section 2.4.1 for the heat supply components. Due to the efficiency advantage, the heat pump is used first, then the electric boiler, and finally, the heat storage:

$$\dot{Q}_{HP}^{\text{set}} = \dot{Q}_{HD}, \quad (24)$$

$$\dot{Q}_{EB}^{\text{set}} = \dot{Q}_{HD} - \dot{Q}_{HP}, \quad (25)$$

$$\dot{Q}_{HS}^{\text{set}} = -(\dot{Q}_{HD} - \dot{Q}_{HP} - \dot{Q}_{EB}). \quad (26)$$

The battery storage balances the electric power between renewable supply and electrical consumption:

$$P_B^{\text{set}} = P_{PV} - P_{ED} - P_{HP} - P_{EB}. \quad (27)$$

4.2.2. Strategy II: Advanced rule-based

The advanced rule-based dispatch strategy employs a state machine framework to optimize the utilization of components, focusing on thermal-driven components. The electric-driven components are controlled via the above-mentioned priority order list (cp. Eq. (27)).

The strategy defines three states Z_j for each flexible component (0: Off, 0.5: Part load, 1: Full load), such as the heat pump and the electric boiler in the heating supply. The changes between these states are governed by temperature limits ($T_j^{\text{on}}, T_j^{\text{off}}$) and transition criteria outlined in Table 5.

The state then determines the component power setpoint:

$$\dot{Q}_j^{\text{set}} = \begin{cases} \dot{Q}_j^{\text{max}} & \text{if } Z_j = 1 \\ \dot{Q}_j^{\text{max}} \cdot x_{PL,j} & \text{if } Z_j = 0.5 \\ 0 & \text{if } Z_j = 0 \end{cases} \quad (28)$$

where the part load factor $x_{PL,j}$ is determined through the following equation:

$$x_{PL,j} = 1 - 0.5 \cdot \frac{T_{HS} - T_j^{\text{on}}}{T_j^{\text{off}} - T_j^{\text{on}}}. \quad (29)$$

The heat set point of the heat storage tank then results from the difference between the heat supply and demand:

$$\dot{Q}_{HS}^{\text{set}} = \dot{Q}_{SH} - \dot{Q}_{HD} + \dot{Q}_{HP} + \dot{Q}_{EB}. \quad (30)$$

This strategy defines the temperature limits for the transition criteria ($T_j^{\text{on}}, T_j^{\text{off}}$) as optimization variables in addition to the design optimization variables (cf. Eq. (A.1)).

4.2.3. Strategy III: Optimal dispatch

Analogous to Section 2.4.3, the optimal dispatch problem for calculating the optimal trajectory of the component power is introduced below.

$$\underset{P_i, Q_j}{\text{minimize}} \quad J_0 \quad \forall i \in \mathbb{I}_e \wedge j \in \mathbb{I}_h \quad (31)$$

$$\text{subject to} \quad \dot{Q}_i^{\text{min}} \leq \dot{Q}_i(t) \leq \dot{Q}_i^{\text{max}} \quad \forall i \in \mathbb{I}_h, \quad (32)$$

$$T_{HS}^{\text{min}} \leq T_{HS}(t) \leq T_{HS}^{\text{max}} \quad \forall t, \quad (33)$$

$$\text{eqs. (2) – (15)} \wedge \text{(22) – (23)} \wedge \text{(A.2) – (A.5)}$$

$$\wedge \text{(A.7)} \wedge \text{(A.9) – (A.11)}$$

$$\text{with} \quad \mathbb{I}_e = \{B, EG\} \wedge \mathbb{I}_h = \{HP, EB, HS\}.$$

Table 6

(a) Component dimensioning S_i and the associated TAC of the considered energy system using the different dispatch strategies (DS) in the design phase (DP). (b) The TAC resulting from applying the different dispatch strategies on the pre-designed systems in the operating phase (OP).

(a) Design phase (DP)			
	Dispatch strategy in DP		
	I	II	III
Dimensioning			
P_{PV}^* in kW	245.54	250.35	272.85
C_B^* in kWh	124.80	122.12	117.72
$\dot{Q}_{HP}^*$ in kW	55.38	39.03	45.22
$\dot{Q}_{EB}^*$ in kW	58.67	8.86	0
V_{HS}^* in m ³	0	3.51	3.75
TAC in T€/a	48.81	47.62	44.64
(b) Operating phase (OP)			
	Dispatch strategy in DP		
	I	II	III
	TAC in T€/a		
DS in OP			
I	48.81	47.92 ^a	48.25 ^a
II	48.81	47.62	49.20
III	48.80	45.12	44.64

^a Infeasible solution.

The optimization determines the battery and grid supply power (P_B , P_{EG}), as well as the heat pump, electric boiler, and heat storage power ($\dot{Q}_{HP}$, $\dot{Q}_{EB}$, $\dot{Q}_{HS}$). The objective of the optimization problem is the minimization of annual operation costs. In addition to the limitation of the electrical components (cf. Section 2.4.3), the power of the heat supply component (cf. Eq. (32)) and the temperature of the heat storage (cf. Eq. (33)) are also limited.

4.3. Results for multi-energy system

First, the results of the system design optimization (design phase) using the presented dispatch strategies are discussed. Then, the implications of using a different dispatch strategy in the operating phase than in the design phase are analyzed.

Design phase - Table 6 (a) shows the results of the optimal dimensioning and the associated TAC applying the different dispatch strategies. As expected, the best objective value results from the optimal dispatch (Strategy III). As all flexible components are used optimally, the photovoltaic system is designed to be the largest; it is worth installing both storage systems, and no inefficient electric boiler is required for the heat supply. A smaller heat pump and an additional electric boiler are installed with rule-based dispatch strategy II. This is because the heat pump is not operated using the surplus electricity from the photovoltaic system and does not depend on the COP in Dispatch Strategy II. As a result, the operation of the heat pump is not as economical as with optimal dispatch, and it is more worthwhile installing an additional electric boiler (low investment costs; high operating costs) instead of a larger heat pump (high investment costs; low operating costs). Dispatch strategy I leads to the worst result. Here, the heat-led components are controlled directly by the heat demand, and the heat storage is not used as buffer storage. For this reason, heat storage is not worthwhile with this dispatch strategy, and a large heat pump and electric boiler must be installed instead.

In conclusion, this case study shows that the choice of dispatch strategy has a significant impact on the optimal dimensioning and also, in particular, on the selection of components.

Operating phase - Table 6 (b) shows the TAC of the pre-designed systems using the different dispatch strategies in the operating phase.

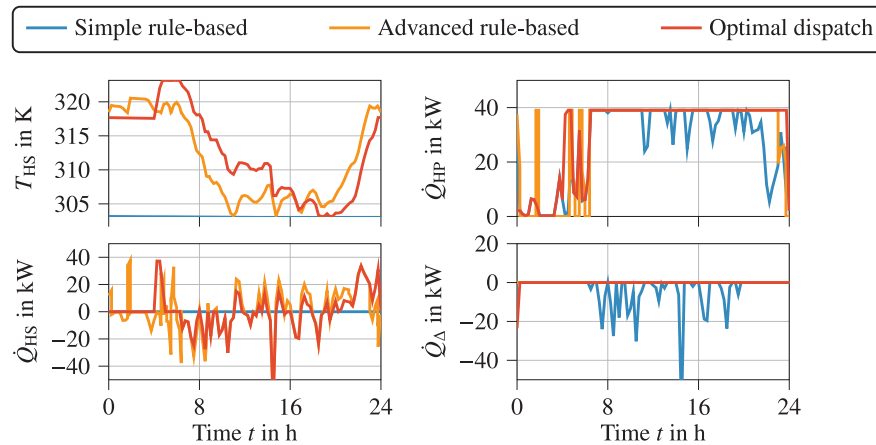


Fig. 6. The resulting heat storage temperature T_{HS} and heat power $\dot{Q}_{HS}$, the heat power of the heat pump $\dot{Q}_{HP}$, and the heat bus power difference $\dot{Q}_A$ apply the different dispatch strategies in System Design II.

Analyzing the table column by column shows how much more expensive the planned system will be if a different dispatch strategy is used in the operating phase than in the design phase.

As no heat storage is installed, there is no flexibility in the heat supply in System Design I. As a result, all dispatch strategies operate the components likewise, resulting in similar TACs. Only optimal dispatch performs slightly better, as self-discharge losses can be reduced here due to the assumption of perfect forecast (see explanation in Section 3.2).

For System Design II, different results are found. As expected, the cost of the optimal dispatch is lower due to the better operation of the dispatchable components based on the assumed perfect forecast. Applying the simple rule-based dispatch strategy (Dispatch Strategy I) in System Design II leads to an infeasible solution because the heat demand cannot be satisfied in every time step. The simple rule-based dispatch strategy does not use heat storage to buffer the heat demand. As a result, the heat storage may be empty at peak load times, and the heat pump, including the electric boiler, may need to be oversized to meet the maximum heat load. This phenomenon can be seen in Fig. 6. The figure shows the course of the temperature and the heat output of the heat storage, the heat output of the heat pump, and the sum of the power difference of the heat bus ($\dot{Q}_A = \sum_{j \in J_h} \dot{Q}_h$) for period 6. If the sum of the power differences $\dot{Q}_A$ is less than zero, the heat requirement cannot be met at this time. This can be seen for Dispatch Strategy I during the day from 6–20 h. Since the heat storage is not charged in Dispatch Strategy I, the heat storage cannot be used during the day to meet the high heat demand ($\dot{Q}_{HD} > \dot{Q}_{HP}^{rated}$), unlike the other dispatch strategies. The same effect can be observed with System Design III. Dispatch Strategy I cannot always cover the heat demand, which leads to an infeasible solution. Applying Dispatch Strategy II in System Design III leads to an acceptable solution, but the cost is 10.21 % higher than the planned system cost using the optimal dispatch.

If Table 6 (b) is analyzed column by column, the effect can be analyzed by what percentage the system would be cheaper if it had been designed directly with the dispatch strategy also used in the operating phase. Dispatch strategy I can only be used if the system has also been designed with this dispatch strategy. Otherwise, the heat supply system must be designed to be more significant. With Dispatch Strategy II, the system costs are 2.50 % higher if System Design I is used and 3.32 % higher if System Design III is used. This means the system is up to 3.32 % more expensive if designed with optimal dispatch, but the rule-based dispatch strategy is used during operation. With Dispatch Strategy III, the costs are 9.31 % and 1.06 % higher if the system is designed with a rule-based dispatch strategy.

The case study shows that the application of different dispatch strategies in the design phase not only leads to different component

dimensioning but also differs in the selection of components. For example, applying Dispatch Strategy I, no heat storage is installed, but an electric boiler with Dispatch Strategy III is the opposite. Furthermore, assuming a dispatch strategy in the design phase that is not applied in the operating phase leads to system designs where the energy demand cannot be met at any time. In particular, if optimal dispatch is assumed in the design phase, required components must be suitably oversized, or additional backup components must be installed. Otherwise, rule-based dispatch can lead to an unreliable energy supply.

5. Conclusion

This paper examines the effects of dissimilar dispatch strategies in decentralized energy systems' design and operating phases. Two energy systems, one simple and one multi-energy system, are explored applying two rule-based dispatch strategies and the optimal dispatch approach for each.

In the design phase, the results vary depending on the case study. Case study 1b considers a very simplified scenario (no demand charge and no heat supply). In this case, the choice of dispatch strategy does not influence the optimal system design. In case studies 1a and 2, considering more complex scenarios, the choice of dispatch strategy clearly influences the system operation performance, the design of the components, and even whether a component should be installed or not. The optimal dispatch consistently outperforms the rule-based strategies because it assumes perfect forecasting and considers the entire system of equations. This means, for example, that the storage systems, in particular, are designed with a higher capacity when optimal dispatch are used. In the heat supply system – without reliable infrastructure – considering optimal dispatch, the heat pump is dimensioned more diminutive, and the electric boiler is not even installed. There are also differences between the two rule-based dispatch strategies. In case study 1a, the battery storage system is designed larger with the advanced rule than with the simple one, and in case study 2, different storage systems are installed when applying the two rule-based dispatch strategies.

In the operating phase, if the designed energy systems are operated with a different dispatch strategy, there is also a significant effect. Depending on the case study, the results are more evident. Again, no significant influence can be seen for case study 1b. For case study 1a, on the other hand, the costs are up to 10.54 % higher if the optimal dispatch is applied in the design phase and a rule-based dispatch strategy is used in the operating phase. At the same time, the cost would be reduced by up to 6.07 % if the rule-based dispatch strategy, which is also used in the operating phase, was directly selected in the design phase.

For the multi-energy system, this effect is even more remarkable. Unlike the electrical supply, the heat supply system does not contain a reliable source. As a result, if the components are undersized, a reliable heat supply may not be guaranteed. For example, the simple rule-based dispatch strategy cannot guarantee a reliable heat supply if the system has been designed using the optimal dispatch or the advanced rule-based dispatch. As in Case Study 1a, the planned costs will also increase (10.21 %) if the system is designed with optimal dispatch and operated with the advanced rule-based dispatch. In addition, the system would have been 3.32 % cheaper if the advanced rule-based dispatch had been used in the design phase.

It is common practice in energy system analysis to assume an optimal dispatch when designing the system. However, caution is advised if the system designed with optimal dispatch cannot be realized. Planned system costs can increase significantly, and the system dimensioning and selection may not be optimal. This study has shown that system costs can increase by up to 10.54 % if the dispatch strategy used in the design phase is not applied during operation. Conversely, the system could have been designed more cost-effectively, with potential savings of up to 6.07 %, if the rule-based dispatch strategy had been used directly in the design phase. For energy systems with unreliable infrastructure, such as heat demand, components must be sufficiently oversized or include backup systems to ensure a reliable energy supply.

For engineers and planners, the following recommendations can be drawn from this study:

- Prioritize the alignment of dispatch strategies between the design and operating phases to minimize cost increases and ensure optimal system performance.
- Explore and invest in advanced dispatch methodologies that more closely mimic optimal dispatch, thereby reducing the performance gap observed when transitioning from design to operation.
- Ensure sufficient component oversizing or backup systems in scenarios where the infrastructure reliability is a concern to guarantee uninterrupted energy supply.

Rule-based dispatch strategies can be easily integrated into optimization models and real applications. However, the formulation considering multiple components, especially in the design phase (when component sizing must be done), can become complex. In addition, this study shows that their performance can deviate significantly from optimal dispatch strategies that require perfect predictions and extensive system knowledge. These deviations highlight the limitations of rule-based approaches. Modern dispatch approaches, such as model predictive control or machine learning techniques, may reduce these deviations by providing more adaptive and responsive strategies. Future work should quantify the impacts of these advanced dispatch strategies in both the design and operating phases to understand their potential benefits and limitations better.

In addition to considering alternative dispatch strategies, further real-world limitations and uncertainties, such as the uncertainty of future load and generation profiles in the operating phase, component degradation, model inaccuracies, or safety margins, were neglected here to isolate the effects of dispatch strategies.

To strengthen the conclusions of this study, future research should aim to validate the model and results with experimental data or real-world case studies. Such validation would provide empirical evidence of the effectiveness of different dispatch strategies and highlight any discrepancies between theoretical predictions and actual performance.

CRediT authorship contribution statement

Lukas Koenemann: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Astrid Bensmann:** Supervision,

Methodology, Funding acquisition, Conceptualization. **Richard Hanke-Rauschenbach:** Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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The results were achieved by computations on the cluster system at the Leibniz Universität Hannover, Germany.

Appendix A. Design optimization, additional component models, and parameters for case study 2

System design optimization - The problem formulation is similar to case study 1:

$$\underset{S_i, F_{B,start}, T_{HS,start}}{\text{minimize}} \quad J_c + J_m + J_o \quad \forall t, \quad (\text{A.1})$$

$$\sum_{i \in \mathbb{I}_h} f_{h,i} \cdot \dot{Q}_i(t) = 0 \quad \forall t, \quad (\text{A.2})$$

$$T_{HS,end} \geq T_{HS,start}, \quad (\text{A.3})$$

$$\text{eqs. (2)–(33)} \wedge (\text{A.4})–(\text{A.13})$$

with

$$\mathbb{I}_h = \{\text{HD, HP, EB, HS}\},$$

The objective of the optimization problem is also to minimize TAC. In addition to the optimization variables already mentioned in Section 2.2, the rated power of the heat pump $\dot{Q}_{HP}^r$, the electric boiler $\dot{Q}_{EB}^r$, the volume of the heat storage V_{HS} , and the initial temperature of the heat storage $T_{HS}(t_{start})$ are also optimized (S_i). The constraints from the optimization problem in Case Study 1 also apply to this optimization problem. In addition, the constraint for maintaining the continuous power balance (cf. Eq. (A.2)) and the cyclical boundary condition for the heat storage (cf. Eq. (A.3)) are introduced. In addition, compliance with the system component models and the selected dispatch strategy must be given. The optimization is performed using the same optimization approach as in the previous case study.

Non-dispatchable components - Fig. A.7 shows the typical periods for the input profiles of case study 2. Four input profiles are required:

- Electrical demand (P_{ED})
- Heat demand ($\dot{Q}_{HD}$: hot water & space heating demand)
- Normalized photovoltaic power (u_{PV})
- Ambient temperature (T^{amb})

The method of Pfenninger and Staffell (2016) is used to calculate the normalized photovoltaic power and the belonging temperature profile. The energy demand profiles are also calculated through an integrated thermal–electrical demand model (McKenna and Thomson, 2016).

Heat pump model - The coefficient of performance (COP) describes the relationship between the required electrical power and the heat output:

$$\eta_{COP,HP} = \frac{\dot{Q}_{HP}}{P_{HP}}. \quad (\text{A.4})$$

According to Ertesvåg (2011), the COP

$$\eta_{COP,HP} = \epsilon_{HP} \cdot \left(1 - \frac{T^{amb}}{T^{sup}}\right)^{-1} \quad (\text{A.5})$$

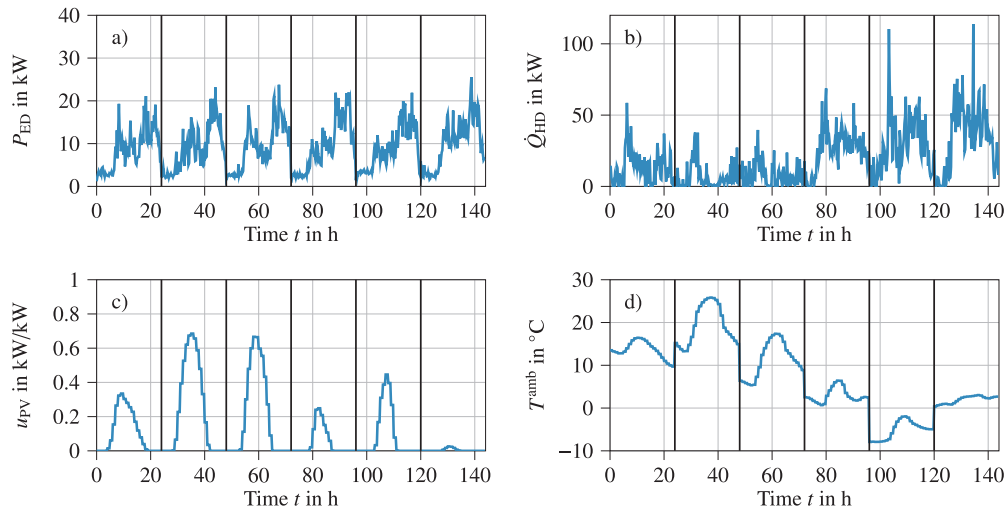


Fig. A.7. The original and the aggregated time series for case study 2: (a) Electric demand, (b) heat demand, (c) normalized photovoltaic power, and (d) ambient temperature. A vertical black line separates the six typical days.

can be calculated by the efficiency factor ϵ_{HP} (depending on the type of heat pump) and the inverse Carnot efficiency. The ambient temperature T^{amb} is provided as an input, and the supply temperature is assumed to be constant ($T^{sup} = 325$ K). Considering the operating limits of the heat pump, the heat output can be calculated using the following set of rules.

$$\dot{Q}_{HP} = \begin{cases} \dot{Q}_{HP}^r & \text{if } \dot{Q}_{HP}^{set} > \dot{Q}_{HP}^r \\ \dot{Q}_{HP}^{set} & \text{if } 0 \leq \dot{Q}_{HP}^{set} \leq \dot{Q}_{HP}^{rated} \\ 0 & \text{otherwise} \end{cases} \quad (A.6)$$

The dispatch strategy calculates the heat pump setpoint $\dot{Q}_{HP}^{set}$.

Electric boiler model - A constant efficiency factor η_{EB} is assumed to calculate the boiler's power consumption (Wilke et al., 2018):

$$P_{EB} = \frac{\dot{Q}_{EB}}{\eta_{EB}}. \quad (A.7)$$

Considering the maximum heat flow, the heat output results from the following rule:

$$\dot{Q}_{EB} = \begin{cases} \dot{Q}_{EB}^r & \text{if } \dot{Q}_{EB}^{set} > \dot{Q}_{EB}^r \\ \dot{Q}_{EB}^{set} & \text{if } 0 \leq \dot{Q}_{EB}^{set} \leq \dot{Q}_{EB}^r \\ 0 & \text{otherwise.} \end{cases} \quad (A.8)$$

Heat storage model - The heat storage tank is modeled as a perfectly stratified tank with a uniform temperature distribution (Vandewalle and D'haeseleer, 2014). The energy balance determines the temperature of the storage water T_{HS} :

$$V_{HS} \cdot \rho_w \cdot c_{p,w} \cdot \frac{dT_{HS}}{dt} = A_{HS}^{sur} \cdot U_{l,HS} \cdot (T^{amb} - T_{HS}) + \dot{Q}_{HS} \cdot \begin{cases} \eta_{HS} & \text{if } \dot{Q}_{HS} \geq 0 \\ \frac{1}{\eta_{HS}} & \text{if } \dot{Q}_{HS} < 0, \end{cases} \quad (A.9)$$

where ρ_w and $c_{p,w}$ are the density and specific heat capacity of the storage water, V_{HS} and A_{HS}^{sur} are the volume and the surface area of the storage tank, $U_{l,HS}$ is the heat loss coefficient, and η_{HS} is the charging respectively discharging efficiency of the heat storage. The storage volume

$$V_{HS} = \pi \cdot r_{HS}^2 \cdot h_{HS}, \quad (A.10)$$

and surface area

$$A_{HS}^{sur} = 2 \cdot \pi \cdot r_{HS} \cdot (r_{HS} + h_{HS}), \quad (A.11)$$

are calculated with radius r_{HS} and height h_{HS} , assuming a cylindrical tank. An aspect ratio of $h_{HS}/r_{HS} = 1$ is chosen to minimize the surface

Table A.7

Economic parameters of case studies 1a, 1b, and 2 for the selected components: Battery (B), photovoltaic system (PV), heat pump (HP), electric boiler (EB), and heat storage (HS).

	c^{acq}	c^m in %	T_i in a	Source
B	500 €/kWh	2	15	Kost (2021)
PV	1500 €/kW	2	30	Kost (2021)
HP	629 €/kW	1.5	18	Ministerium für Umwelt, Klima und Energiewirtschaft Baden-Württemberg (2023)
EB	362 €/kW	1	30	Ministerium für Umwelt, Klima und Energiewirtschaft Baden-Württemberg (2023)
HS	1880 €/m ³	1.3	20	Ministerium für Umwelt, Klima und Energiewirtschaft Baden-Württemberg (2023)
	c^{power} in €/kW a	c^{energy} in ct/kWh	Source	
1a	102.54	15.32	Bundesverband der Energie- und Wasserwirtschaft (2021), Enercity Netz (2021)	
1b	0	21.19	Bundesverband der Energie- und Wasserwirtschaft (2021), Enercity Netz (2021)	
2	0	48.12	Bundesverband der Energie- und Wasserwirtschaft (2023)	

area of the heat storage (van der Heijde et al., 2019). Depending on the heat flow set point of the dispatch strategy, the heat flow can be calculated using the following rule:

$$\dot{Q}_{HS} = \begin{cases} \dot{Q}_{HS}^{max} & \text{if } Z_{HS} = 0 \wedge \dot{Q}_{HS}^{set} > \dot{Q}_{HS}^{max} \\ \dot{Q}_{HS}^{set} & \text{if } Z_{HS} = 0 \wedge -\dot{Q}_{HS}^{max} \leq \dot{Q}_{HS}^{set} \leq \dot{Q}_{HS}^{max} \\ -\dot{Q}_{HS}^{max} & \text{if } Z_{HS} = 0 \wedge -\dot{Q}_{HS}^{max} > \dot{Q}_{HS}^{set} \\ \dot{Q}_{HS}^{set} & \text{if } Z_{HS} = -1 \wedge 0 \leq \dot{Q}_{HS}^{set} \\ \dot{Q}_{HS}^{set} & \text{if } Z_{HS} = 1 \wedge 0 \leq \dot{Q}_{HS}^{set} \\ 0 & \text{otherwise,} \end{cases} \quad (A.12)$$

where $\dot{Q}_{HS}^{max}$ is the maximum heat flow, and Z_{HS} is the heat storage state calculated by the limits for the storage water temperature (T_{HS}^{min} , T_{HS}^{max}):

$$Z_{HS} = \begin{cases} 1 & \text{if } T_{HS} > T_{HS}^{max} \\ 0 & \text{if } T_{HS}^{min} \leq T_{HS} \leq T_{HS}^{max} \\ -1 & \text{if } T_{HS} < T_{HS}^{min}. \end{cases} \quad (A.13)$$

Table A.8

Physical parameters for the selected components: Battery (B), photo-voltaic system (PV), heat pump (HP), electric boiler (EB), and heat storage (HS).

Parameter	Value	Source
PV azimuth	α_{PV} 180 °	own asm.
PV tilt	γ_{PV} 35 °	own asm.
PV loss factor	l_{PV} 0.1	Pfenninger and Staffell (2016)
B self dis. rate	k_B^{sd} 0.1%/d	Tiemann et al. (2020)
B efficiency	η_B 0.9487	Tiemann et al. (2020)
B Specific pow.	Π_B 1 $\frac{kW}{kWh}$	Tiemann et al. (2020)
HP efficiency	ϵ_{HP} 0.335	own asm.
EB efficiency	η_{EB} 0.99	Wilke (2020)
HS efficiency	η_{HS} 0.96	Elsner (2015)
Heat loss coef.	$U_{l,HS}$ 0.35 $\frac{W}{m^2K}$	Reflex Winkelmann GmbH (2019)

Table B.9

Calculated weight factors w_p for case studies 1 a, 1 b (cf. Section 3) and 2 (cf. Section 4.3).

Period	w_p for period p					
	1	2	3	4	5	6
Case study 1	132	132	84	26	–	–
Case study 2	63	70	87	125	10	9

System parameter - Table A.7 lists the economical parameters of the case studies and Table A.8 lists the physical parameters of the components.

Appendix B. Weight factors

Table B.9 shows the calculated weight factors w_p for each typical period p of the three case studies.

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